

Levi Mackay

Computer Science Student, Software Engineer

Rexburg, ID · levimackay@gmail.com · 208.284.4004 · levimackay.com · github.com/levimackay · linkedin.com/in/levi-mackay-217380396

Built Izvor, my own programming language, and its compiler in C. Typed source goes through a lexer, parser, type checker, and C code generator and comes out a native binary, with no dependencies and no LLVM.

OPEN SOURCE PROJECTS

izvor, a programming language and compiler in C

github.com/levimackay/izvor · MIT · C11, no dependencies

- Hand written lexer and recursive descent parser producing a tagged union syntax tree, feeding a type checker, an interpreter, and a C code generator that builds native binaries through clang. C11, no libraries, no parser generator, no LLVM.
- 36 golden programs must match byte for byte between the interpreter and the compiled binary, errors included. A fixed seed fuzzer pushes 20,000 inputs through the front end, and CI runs the suite on Linux and macOS under sanitizers.
- Published the build as a free line by line course at tutorial.levimackay.com so other people can follow it. I used AI assistance to build the course site, and a chapter only goes up once the code behind it is in the repository and passing its tests.

Lydia (CLI), a local AI coding agent

github.com/levimackay/lydia-cli · MIT · 36 stars, 8 forks

- A terminal coding agent that reads, edits, and tests code entirely against local Ollama models. No API keys, and nothing leaves the machine. Roughly 7,100 lines of Python across 75 modules.
- 475 tests across 56 files run in GitHub Actions on every push, and I do not merge anything until the suite is green.
- Two developers found the project on their own and opened pull requests that I reviewed and merged: Linux systemd user service support, and a fix for boolean config values stored as strings.

EXPERIENCE

Main Street Sites, Founder and Developer

Rexburg, ID · May 2026 to Present

- Built and shipped the studio itself on React 19 and Vite with 32 tests in CI, including the pricing, the intake form, and seven demo builds. Forms route through Web3Forms, so there is no server to keep alive.
- Set the pricing, wrote the outreach, and pitched local business owners cold. Five paying customers so far, with no public checkout by design: a payment link goes out by hand after a conversation.

Sandbox at BYU-Idaho, Fellow

Rexburg, ID · Sep 2026 to Present

Selected in April 2026 for Sandbox, a venture program where students spend the academic year building a tech startup.

BYU-Idaho, CS Teaching Assistant and CSE Employability and Outreach Assistant

Rexburg, ID · Apr 2026 to Present

Work with about 100 students a semester on programming fundamentals and debugging strategy, mostly by asking the next question instead of handing over the answer, and grade assignments with written feedback. Also work with CS faculty on internship promotion, employer visits, and technical workshops.

SafeStreets, Dealer Operations and IT Specialist

Rexburg, ID · Mar 2025 to Aug 2025

Managed dealer account data and system permissions, and resolved record discrepancies several teams depended on.

PROJECTS

RexNest, off campus housing marketplace

Next.js 16, Supabase Postgres, Stripe

- Ranks apartments on a weighted score over price, distance to campus, amenities, and resident reviews. Listings with fewer than three reviews fall back to the site average, so a place cannot ride on one or two ratings. 23 Postgres tables behind 129 lines of row level security policy, and 123 tests in 10 files.
- Monetization built on Stripe: recurring subscriptions for property managers and one time paid boosts for listings. Billing tables are writable only by the Stripe webhook.

Serbo-Croatian dictionary and language reference

Original research · Python, Typst

Built a Python and Typst pipeline that merges, indexes, and typesets 5,709 entries from two years of my own handwritten field notes into a 301 page reference. Anything the pipeline adds or corrects is flagged in the output rather than silently overwriting what I recorded.

EDUCATION

Brigham Young University-Idaho

Sep 2024 to Dec 2027 (expected)

B.S. Computer Science, Minor in AI Engineering · 3.98 GPA

SKILLS

Programming: C, Python, TypeScript, JavaScript, SQL, C# · **Web:** HTML, CSS, React, Next.js, Tailwind, three.js
Data: PostgreSQL, Supabase · **Tooling:** Git, GitHub Actions, clang, pytest, Cloudflare · **Spoken:** Serbian, Croatian, Bosnian